

VDraw

User Manual

Two-dimensional vector drawing and technical illustration

This manual describes the complete VDraw interface: menus, toolbars, drawing tools, the properties panel, import and export.

Contents

Contents 2
1. Introduction 5
1.1 Key ideas 5
1.2 Files VDraw works with 5
2. The main window 6
2.1 What the pointer does on the canvas 7
3. Documents and pages 8
3.1 Creating, opening and saving 8
3.2 Page setup 8
Page properties 8
3.3 Several pages 9
4. The toolbars button by button 10
4.1 Main toolbar 10
4.2 Document toolbar 11
4.3 Object toolbar 11
5. Drawing shapes 12
5.1 Straight lines 13
5.2 Curves 13
5.3 Rectangles 14
5.4 Circles 14
By centre and radius 14
By three points 15
5.5 Ellipses 15
5.6 Polygons 16
Regular polygon 16
Free polygon 16
5.7 Text 17
5.8 Images 18
5.9 Dimensions 19
6. Editing curves 20
6.1 Entering node editing 20
6.2 Working with nodes 20
6.3 The node toolbar 21
6.4 A worked example 21
7. Boolean operations 22
7.1 The common sequence 22
7.2 The five operations 22
Union — Ctrl+Alt+U 22
Subtract — Ctrl+Alt+S 22
Intersect — Ctrl+Alt+I 22
Exclude — Ctrl+Alt+X 23
Combine — Ctrl+Alt+C 23
8. Cutting shapes 24
8.1 Cutting with a straight line 24
8.2 Cutting with a curve 24
8.3 Closing the cut edges 25
9. Line anchors 26
9.1 Attaching an end 26
9.2 The Tangent switch 26
9.3 Releasing an end 26
10. Grouping 27
10.1 Grouping 27

10.2 Ungrouping ... 27
10.3 What a group does to properties ... 27
11. Rulers, guides, grid and snapping ... 28
11.1 Rulers ... 28
11.2 Creating a guide ... 28
11.3 Moving and removing a guide ... 28
11.4 Rotating a guide ... 28
11.5 The grid ... 28
11.6 Snapping ... 29
11.6 Snapping by a chosen node ... 29
12. Custom shapes and .vdrs files ... 30
12.1 Where custom shapes come from ... 30
12.2 Creating a custom shape ... 30
12.3 Inserting a custom shape ... 30
12.4 Importing .vdrs files ... 30
Removing a set ... 31
12.5 The same import from the toolbar ... 31
13. Image editing ... 32
13.1 Entering and leaving image mode ... 32
13.2 The image toolbar ... 32
13.3 Retouching: a worked sequence ... 32
13.4 Working on part of the image ... 33
13.5 Filling an area ... 33
14. The properties panel ... 34
14.1 Several shapes at once ... 34
14.2 General ... 34
14.3 Line ... 35
14.4 Fill ... 36
None ... 36
Solid ... 36
Linear gradient ... 37
Radial and rectangular gradients ... 37
Pattern ... 38
Texture ... 39
14.5 Shadow ... 39
14.6 Transparency ... 40
14.7 Text ... 41
14.8 Link ... 41
15. Export and printing ... 42
15.1 Exporting a raster image ... 42
15.2 Exporting an animation ... 42
15.3 Printing ... 43
16. Animation ... 44
16.1 Making your first animation ... 44
16.2 The toolbar control by control ... 44
16.3 The parameter tree ... 45
16.4 Choosing what Set key records ... 45
16.5 Keys ... 46
What the diamonds mean ... 46
Recording a key ... 46
What happens between keys ... 46
16.6 Selecting keys ... 47
16.7 Moving and deleting keys ... 47
Moving ... 47

Deleting 47
16.8 Moving around the timeline 47
16.9 Keyboard shortcuts in the panel 48
16.10 A worked example: a bouncing ball 48
16.11 Saving and exporting 48
17. Settings 49
17.1 The pages 49
17.2 The buttons 49
18. Keyboard shortcuts 50
19. If something goes wrong 51
19.1 Common situations 51

1. Introduction

VDraw is a two-dimensional vector editor. A drawing is made of shapes — lines, curves, rectangles, circles, polygons, text, images and dimensions — placed on one or more pages of a document. Shapes keep their mathematical description, so they can be moved, scaled and re-edited at any time without losing quality.

This application is a tool that combines vector graphics, simple image editing and animations.

1.1 Key ideas

Five ideas explain most of the program. Reading them first will save time later.

- Document — one .vdr file. It holds every page, every shape, and the patterns and custom shapes used by them.
- Page — one sheet of paper with its own format, orientation, units and background. A document may hold many pages; the tabs at the bottom of the window switch between them.
- Shape — one object on a page. Every shape has an outline (Line), a Fill, a Shadow, a Transparency and a position.
- Units of measure — every number you type in the properties panel is expressed in the units of the current page, chosen in the Units box of the document toolbar. Changing the units converts all existing values.
- Layer — the stacking order. A shape with a higher layer number is drawn over a shape with a lower one.

1.2 Files VDraw works with

Extension	Role
.vdr	VDraw document. The native format; keeps everything.
.vdrs	Custom shape set — one saved group of shapes, ready to be inserted into any drawing.
.vss, .vssx, .vssm	Microsoft Visio stencils. Imported into custom shapes.
.pdf	Imported as pages of shapes.
PNG, JPEG, BMP, SVG	Export targets for the selection.
GIF	Animation export.

2. The main window

The illustration below shows the whole window with every area numbered. The rest of this manual follows the same numbering, so you can always come back here to find where a control lives.

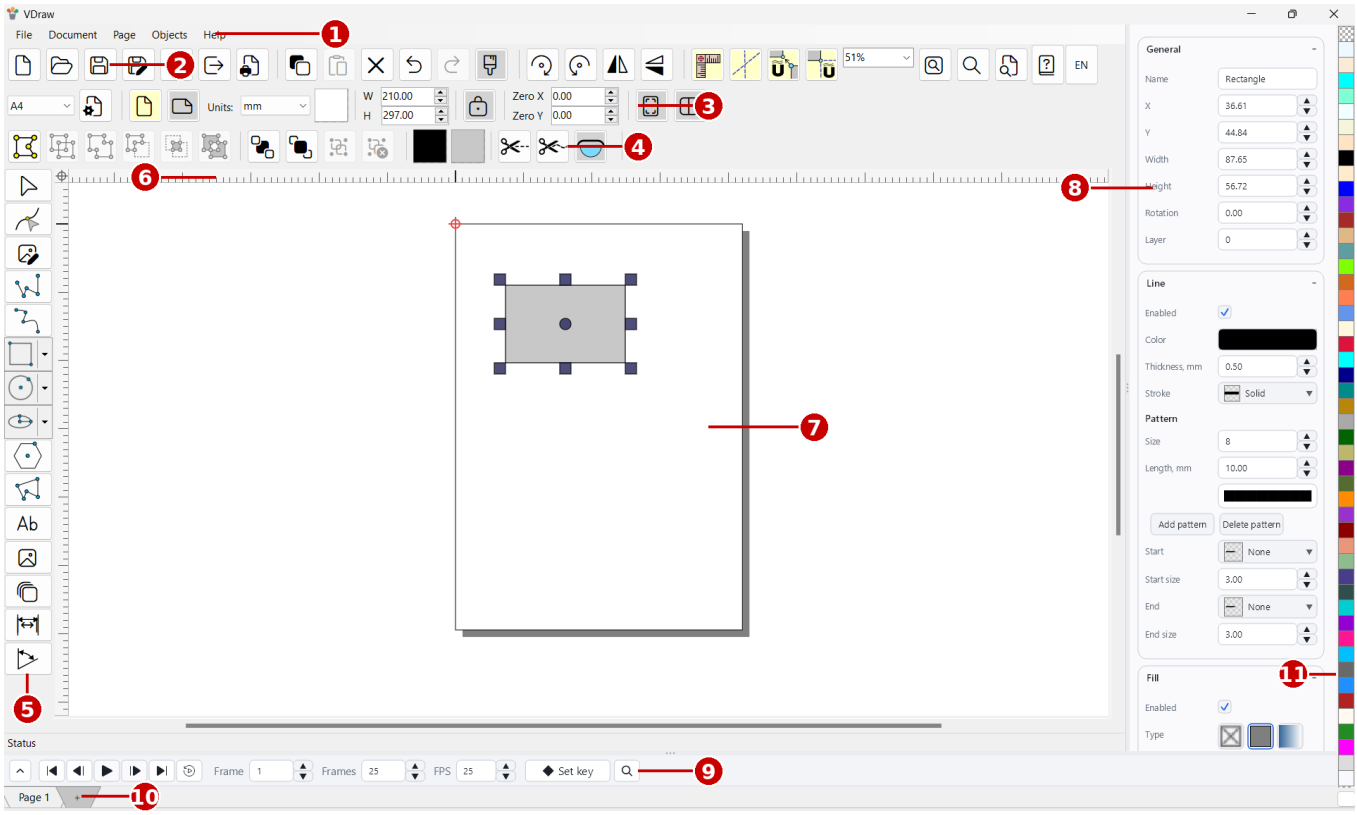


Figure 2.1 — The VDraw main window with its eleven areas

No.	Area
1	Menu bar — File, Document, Page, Objects, Help and Settings.
2	Main toolbar — new, open, save, import, export, print, clipboard, undo and redo, rotation and mirroring, view switches and zoom.
3	Document toolbar — paper format, orientation, units, page width and height, and the page zero point.
4	Object toolbar — grouping, order, boolean operations, node editing, cutting, and the two default colors.
5	Tool palette — the drawing tools: select, lines, curves, rectangles, circles, ellipses, polygons, text, images, dimensions.
6	Rulers — they show the position of the cursor and are the place guides are dragged from.
7	Canvas — the sheet and everything on it.
8	Properties panel — every property of the current selection, grouped into collapsible sections.
9	Animation bar — frames, playback and key setting.
10	Page tabs — switch pages and add a new one with the plus button.
11	Color palette — click a swatch to apply a color to the selection.

2.1 What the pointer does on the canvas

With the Select tool active, the mouse behaves as follows.

Action	Result
Click a shape	Selects it. Handles appear around it.
Click empty space	Clears the selection.
Shift + click	Adds a shape to the selection or removes it.
Drag on empty space	Rubber-band selection of everything the frame touches.
Drag a shape	Moves it. Snapping applies unless Alt is held.
Drag a corner handle	Resizes.
Click a selected shape again	Switches the handles to rotation and skew.
Right-click a selection	Opens the context menu: Copy, Paste, Convert to curves, Convert to bitmap, Group, Ungroup, Delete.
Ctrl + mouse wheel	Zooms around the pointer.
Middle-button drag	Pans the view.

3. Documents and pages

3.1 Creating, opening and saving

The File menu and the left-hand group of the main toolbar carry the document commands.

Command	Shortcut and behaviour
New document	Ctrl+N. Asks for a paper format first, then replaces the current document. Unsaved work is offered for saving.
Open document	Ctrl+O. Opens a .vdr file.
Save	Ctrl+S. Saves to the current file; a new document asks for a name.
Save as	Ctrl+Shift+S. Saves under a new name and continues in the new file.
Import	Ctrl+I. See chapter 10 — it accepts .vdr, .pdf, .vdrs and every Visio format.
Export selection	Ctrl+E. Writes the selected shapes to PNG, JPEG, BMP or SVG.
Print	Ctrl+P.
Close / Exit	Closes the document or the program, offering to save first.

3.2 Page setup

The document toolbar (area 3) sets up the sheet you are drawing on.

1. Choose a paper format in the first combo box — A4, A3, Letter and so on, or Custom.
2. If you chose Custom, type the sheet size into the W and H fields.
3. Press the lock button between them to keep the aspect ratio while you type; with the lock on, changing one side changes the other.
4. Pick portrait or landscape with the two orientation buttons.
5. Choose the units in the Units box. Every number in the properties panel is then expressed in those units.
6. Set Zero X and Zero Y if you want the origin of coordinates somewhere other than the top-left corner of the sheet.

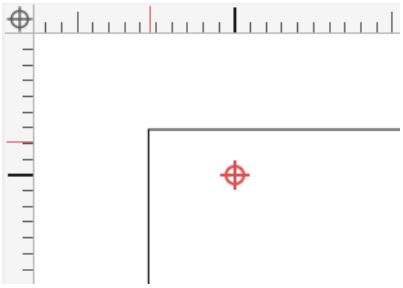


Figure 3.1 — The rulers and the page zero marker (the red cross-hair)

The zero point is drawn on the sheet as a small red cross-hair. All coordinates in the properties panel, and all ruler readings, are measured from it. Moving the zero does not move any shape — only the numbers change.

Page properties

The Page properties button opens a dialog with the same settings plus the page background and name.

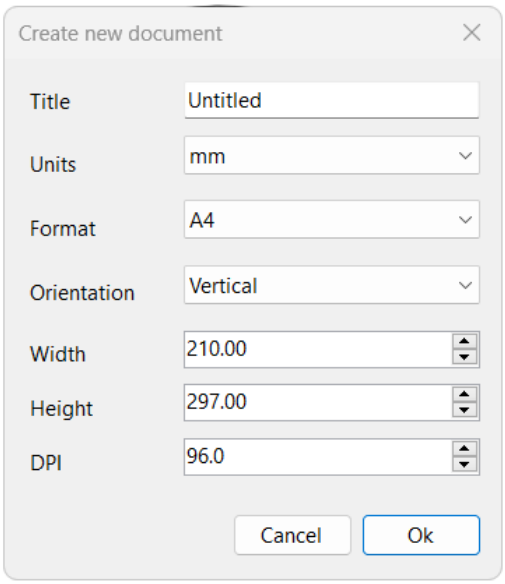


Figure 3.2 — The page properties dialog

3.3 Several pages

- The tabs at the bottom-left switch pages.
- The plus button adds a page after the current one.
- Right-click a tab to rename, duplicate or delete the page.
- Each page keeps its own format, units and background.

4. The toolbars button by button

This chapter lists every button. Within each table the buttons appear in the same left-to-right order as on the screen, so you can follow along with the window in front of you.

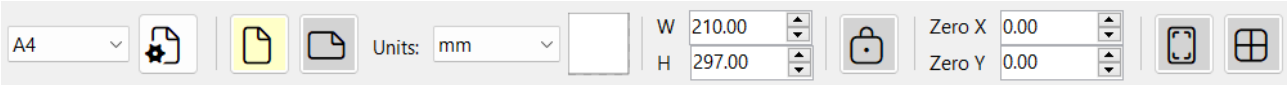
4.1 Main toolbar

Reading left to right, the main toolbar carries seven document commands, the clipboard group, undo and redo, the transform group, the view switches and the zoom controls.



Button	Function
New document	Ctrl+N — asks for a paper format, then replaces the current document.
Open document	Ctrl+O — opens a .vdr file.
Save	Ctrl+S.
Save as	Ctrl+Shift+S.
Import	Ctrl+I — inserts shapes from .vdr, pages from .pdf, and adds .vdrs sets and Visio stencils to custom shapes.
Export selection	Ctrl+E — PNG, JPEG, BMP or SVG.
Print	Ctrl+P.
Copy	Ctrl+C — copies the selected shapes; in image mode copies the pixel selection.
Paste	Ctrl+V.
Delete selected	Del.
Undo	Ctrl+Z.
Redo	Ctrl+Y.
Format painter	Copies the appearance of one shape onto another.
Rotate 90° counter-clockwise	Ctrl+Left.
Rotate 90° clockwise	Ctrl+Right.
Flip horizontally	Shift+H.
Flip vertically	Shift+V.
Rulers	Ctrl+R — shows or hides the rulers.
Guides	Ctrl+; — shows or hides the guide lines.
Snap to objects	Ctrl+Shift+N — hold Ctrl while dragging to suspend it temporarily.
Dynamic guides	Shows alignment lines to nearby shapes while dragging.
Zoom box	Pick a value or type your own. Ctrl+wheel zooms at the cursor.
Zoom to selection	Ctrl+Shift+0.
Zoom to all shapes	Ctrl+0.
Fit page	Ctrl+1.
Help	F1.
Language	Switches the interface language.

4.2 Document toolbar



Control	Function
Paper format	Choose a standard format, or Custom to type your own size.
Page properties	Opens the page dialog: format, size, units and background.
Portrait / Landscape	Sheet orientation.
Units	The unit every number in the program is expressed in.
Page background color	Fills the sheet behind the shapes.
W and H	Sheet width and height in the current units.
Lock aspect ratio	When on, changing one side changes the other proportionally.
Zero X and Zero Y	Position of the origin of coordinates on the sheet.
Printable area borders	Draws the margins your printer cannot reach.
Grid	Ctrl+# — shows the grid.

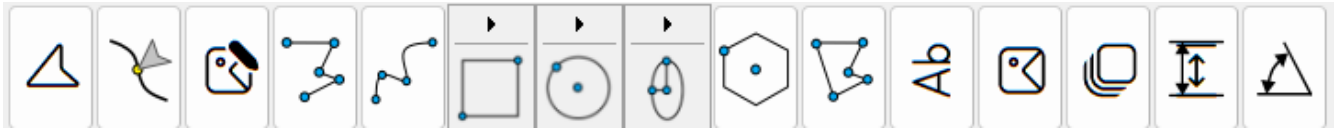
4.3 Object toolbar



Button	Function
Group	Ctrl+G — joins the selected shapes into one group.
Ungroup	Ctrl+Shift+G — breaks the selected group apart.
Bring to front	Ctrl+Shift+Up.
Send to back	Ctrl+Shift+Down.
Union	Ctrl+Alt+U — merges the selected shapes into one.
Subtract	Ctrl+Alt+S — removes the upper shapes from the lower one.
Intersect	Ctrl+Alt+I — keeps only the overlapping area.
Exclude	Ctrl+Alt+X — keeps everything except the overlapping area.
Combine	Ctrl+Alt+C — joins the shapes into one curve keeping separate contours.
Convert to curve	Ctrl+Q — turns a primitive into an editable curve.
Edit nodes	N — shows the nodes and control handles of the selected curve.
Cut with a straight line	K.
Cut with a curve	Shift+K.
Close the cut edges	When on, the parts produced by a cut get a closed contour.
Default outline color	Applies to newly created shapes.
Default fill color	Applies to newly created shapes.

5. Drawing shapes

The tool palette down the left edge holds every drawing tool. A tool stays active until you choose another one or press Esc, which returns to Select.



Tool	Shortcut and note
Select	Esc — pick, move, resize and rotate shapes.
Straight lines	L — click to add nodes, right-click or Enter to finish.
Curve	B — click to add nodes, right-click or Enter to finish.
Rectangle	R — the small arrow opens plain, rounded and chamfered variants.
Circle	C — the arrow switches between by centre and radius, and by three points.
Ellipse	E — the arrow switches between by three points and by five points.
Regular polygon	P — first click sets the centre, second the radius.
Free polygon	Shift+P — click to add vertices, right-click or Enter to close.
Text	T.
Image	Ctrl+Shift+I — places a raster image into a closed contour.
Custom shapes	Inserts a saved set — see chapter 10.
Measure distance	M — three clicks: from, to, and the height of the value label.
Measure angle	Three clicks defining the two arms of the angle.

> *Tools with a small arrow beside them are split buttons. Clicking the button uses the variant shown; clicking the arrow opens the list of variants and remembers your choice.*

5.1 Straight lines

1. Click the Straight lines tool, or press L.
2. Click on the canvas — the first node is placed and a preview segment follows the pointer.
3. Click again for every further node. Each click fixes a segment.
4. Right-click, or press Enter, to finish the line.
5. Press Esc while drawing to abandon the whole shape.

While you click, snapping is active: node positions jump to grid intersections, to guides and to points of other shapes. Hold Ctrl to suspend snapping for a moment.

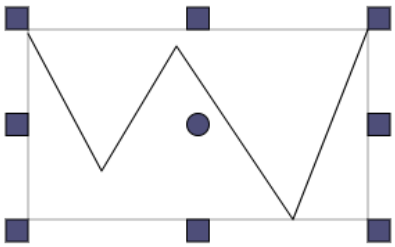


Figure 5.1 — A finished multi-segment line, selected

5.2 Curves

1. Click the Curve tool, or press B.
2. Click to place the first node.
3. Click for each further node — the program passes a smooth curve through all of them.
4. Right-click or press Enter to finish.

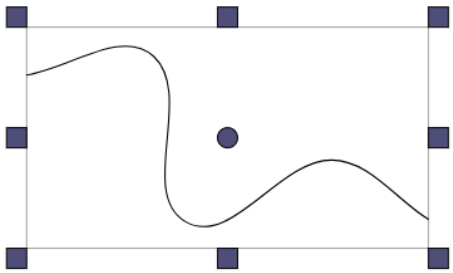


Figure 5.2 — A curve with its nodes

A curve drawn this way can be reshaped afterwards at any time — see chapter 6.

5.3 Rectangles

1. Click the Rectangle tool, or press R. To draw a rounded or chamfered rectangle, click the small arrow first and pick the variant.
2. Press the mouse button at one corner.
3. Drag to the opposite corner; the size is shown as you go.
4. Release to finish.

For rounded and chamfered rectangles the corner size is set afterwards in the General section of the properties panel, in the Round X / Round Y or Chamfer X / Chamfer Y fields.

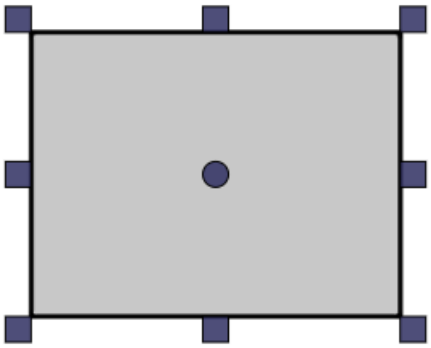


Figure 5.3 — A circle with its centre and radius handles

5.4 Circles

Two construction methods are available from the split button.

By centre and radius

1. Click the Circle tool, or press C.
2. Click at the centre.
3. Move outwards and click again on the radius. The circle is finished.

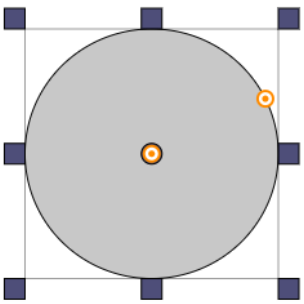


Figure 5.4.1 — A circle with its centre and radius handles

By three points

1. Open the arrow beside the Circle tool and choose the three-point variant.
2. Click the first point on the circumference.
3. Click the second point.
4. Click the third — the only circle through the three points is created.

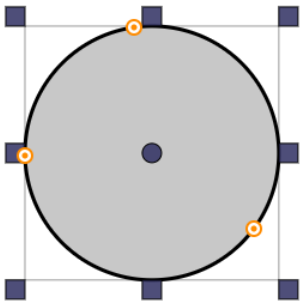


Figure 5.4.2 — A circle with its centre and radius handles

5.5 Ellipses

1. Choose the Ellipse tool, or press E; the arrow selects the three-point or five-point variant.
2. For the three-point variant: click the centre, then a point on the first axis, then a point on the second.
3. For the five-point variant: click five points that the ellipse must pass through.

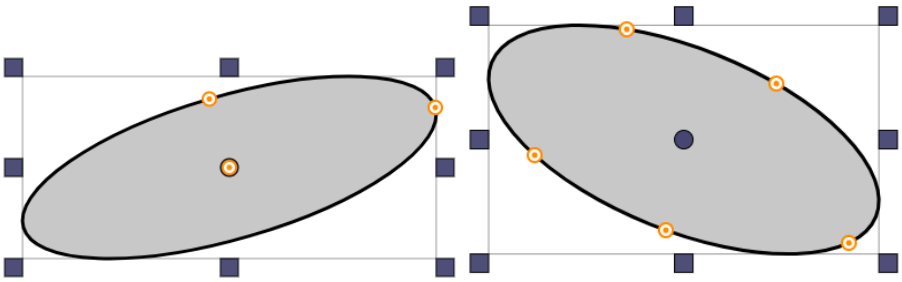


Figure 5.5 — An ellipse with its centre and radii handles, and ellipse built by 5 points.

5.6 Polygons

Regular polygon

1. Click the Regular polygon tool, or press P.
2. Click at the centre.
3. Move outwards and click to set the radius.
4. Set the number of sides afterwards in the Sides field of the properties panel.

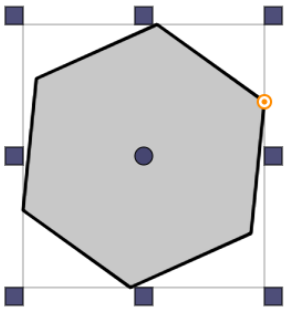


Figure 5.6.1 — A polygon with equal edges built by center and outer point on its radius.

Free polygon

1. Click the Free polygon tool, or press Shift+P.
2. Click each vertex in turn.
3. Right-click or press Enter — the contour is closed automatically between the last vertex and the first.

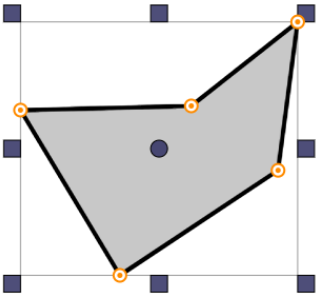


Figure 5.6.2 — A polygon built by multiple points.

5.7 Text

1. Click the Text tool, or press T.
2. Click on the sheet where the text should start.
3. Type. The text appears as you type.
4. Press Esc, or click another tool, to finish.
5. To edit later: select the text shape and change the Text field in the Text section of the properties panel.

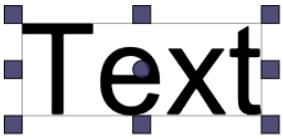


Figure 5.7.1 — A text created at view.

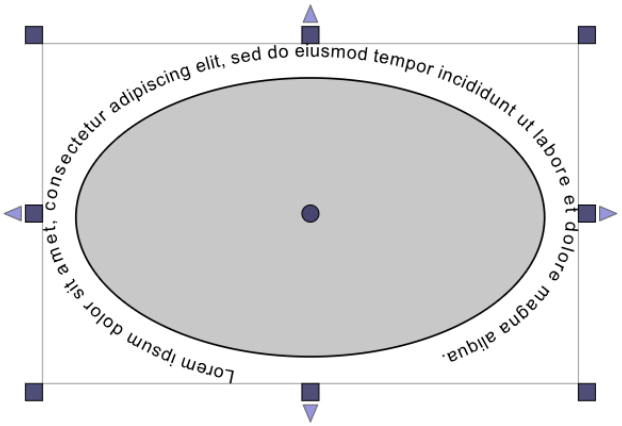


Figure 5.7.2 — A text along other shape path.

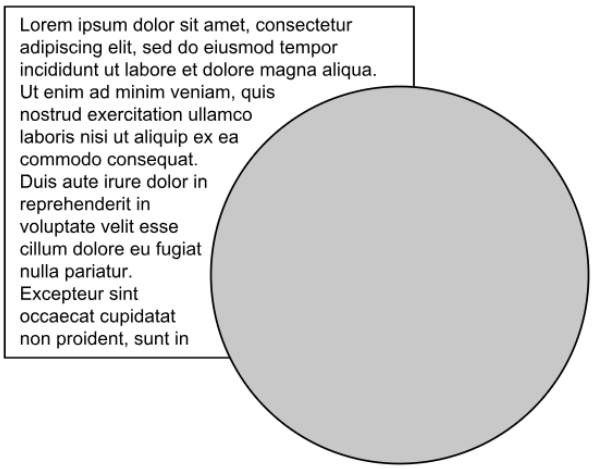


Figure 5.7.3 — A text inside shape, with padding and wrapped by external shape.

5.8 Images

1. Draw a closed shape first — a rectangle, an ellipse or a closed curve. The shape becomes the frame the picture is clipped to.
2. Select that shape.
3. Click the Image tool, or press Ctrl+Shift+I.
4. Choose the picture file. It is placed inside the contour.

An image can also be pasted straight from the clipboard with Ctrl+V; it then arrives in a rectangular frame of its own.



Figure 5.8 — An image loaded and inserted at page.

5.9 Dimensions

1. Click Measure distance, or press M.
2. Click the first point. Ends snap to the points of nearby shapes, so a dimension can be attached to a corner exactly.
3. Click the second point.
4. Move away from the line and click a third time to set how far the value label sits from the measured line.

The measured value is recalculated whenever the dimension or the shapes it is attached to move. To label a drawing that is not to scale, type your own number into the Value field of the Text section — see 14.7.

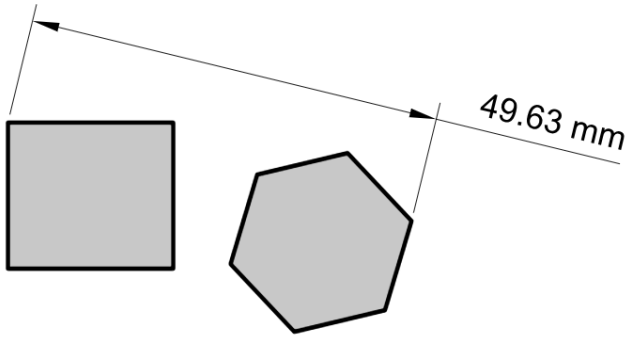


Figure 5.9 — A distance measurement created at page.

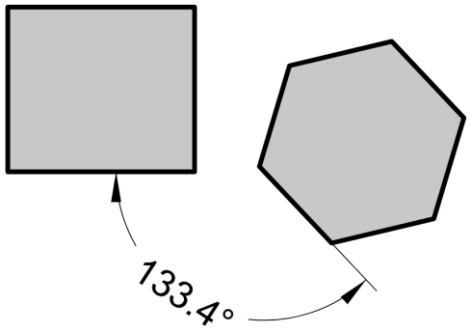


Figure 5.9 — An angle measurement created at page.

6. Editing curves

Every shape in VDraw can become an editable curve, and a curve is edited by moving its nodes and the control handles that shape the segments between them.

6.1 Entering node editing

1. Select the shape with the Select tool.
2. If the shape is a primitive — a rectangle, circle, ellipse or polygon — press Ctrl+Q, or use Convert to curve on the object toolbar. A primitive has no nodes until it is converted.
3. Press N, or click Edit nodes on the object toolbar.
4. The nodes appear as yellow dots and the control handles as blue dots joined to them by thin lines.
5. Press N again, or Esc, to leave node editing.

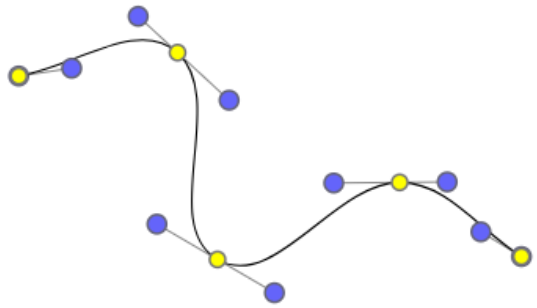


Figure 6.1 — A curve in node editing: yellow nodes, blue control handles

> *Convert to curve cannot be undone by converting back — a rectangle that became a curve stays a curve. Ctrl+Z does reverse it.*

6.2 Working with nodes

Action	How
Select a node	Click it. It is drawn larger while focused.
Select several	Shift + click each one, or drag a frame around them.
Move nodes	Drag any selected node; all selected nodes move together.
Curve a segment	Drag a control handle away from its node. The segment bends.
Add a node	Double-click on the segment where the node should appear.
Delete nodes	Select them and press Del, or use the delete button on the node toolbar.

6.3 The node toolbar

While node editing is active, a small toolbar appears with six commands.

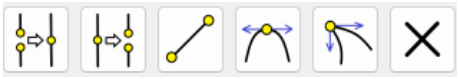


Figure 6.3 — The node editing toolbar

Button	Function
Merge nodes	Ctrl+J — joins two selected ends into one node, closing a gap in the contour.
Split node	Ctrl+Shift+J — breaks the contour at the focused node, turning one contour into two ends.
Straight node	1 — drops the control handles, so both segments meeting at the node become straight.
Smooth node	2 — the two control handles stay collinear, so the curve passes through the node smoothly.
Cusp node	3 — the two control handles move independently, so a corner is allowed.
Delete nodes	Removes the selected nodes and joins the neighbours.

6.4 A worked example

1. Draw a rectangle with R.
2. Press Ctrl+Q to convert it to a curve.
3. Press N — four nodes appear at the corners.
4. Double-click the middle of the top edge to add a fifth node.
5. Drag that node upwards — the top edge becomes a peak.
6. Click the node and press 2 to make it smooth — the peak becomes a rounded hill.
7. Press N to leave node editing.

7. Boolean operations

Boolean operations combine two or more overlapping shapes into a new one. They live in the middle of the object toolbar, and all of them work the same way.

7.1 The common sequence

1. Draw the shapes and arrange them so that they overlap.
2. Select them all — click the first, then Shift + click the others. The order matters for Subtract: the shape selected first, and lying lowest, is the one that is kept.
3. Click the operation on the object toolbar, or press its shortcut.
4. The originals are replaced by a single new shape, which takes the appearance of the bottom shape.

> *Every boolean operation is a single undo step. Ctrl+Z brings back all the original shapes at once.*

7.2 The five operations

Union — Ctrl+Alt+U

Merges everything selected into one outline. Overlaps disappear; the result is the area covered by any of the shapes.

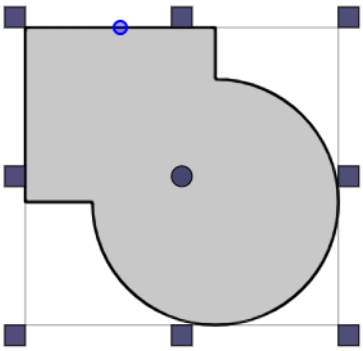


Figure 7.2.1 — Union

Subtract — Ctrl+Alt+S

Removes the upper shapes from the lower one. Use it to punch holes: draw the hole on top of the body, select body first, then the hole, then subtract.

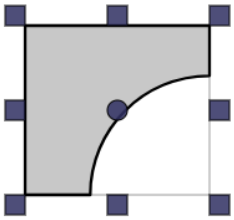


Figure 7.2.2 — Subtract

Intersect — Ctrl+Alt+I

Keeps only the area where the shapes overlap; everything else is discarded.

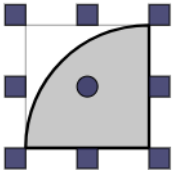


Figure 7.2.3 — Intersect

Exclude — Ctrl+Alt+X

Keeps everything except the overlap — the mirror image of Intersect.

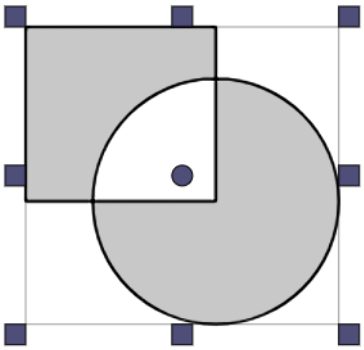


Figure 7.2.4 — Exclude

Combine — Ctrl+Alt+C

Joins the shapes into one curve while keeping their contours separate. Nothing is added or removed geometrically; the shapes simply become one object with several contours. Where contours overlap, the fill leaves a hole, which is the usual way to make a ring or a letter O. The behaviour of overlaps is controlled by the Fill intersection gaps switch in the Fill section.

8. Cutting shapes

The scissors cut a shape into pieces along a line or a curve you draw. Two tools sit at the right of the object toolbar, and a third button decides what happens to the cut edges.

8.1 Cutting with a straight line

1. Select the shape to be cut.
2. Click Cut with a straight line, or press K.
3. Click once where the cut should start — outside the shape is fine.
4. Click a second time where it should end. The shape is divided at once.

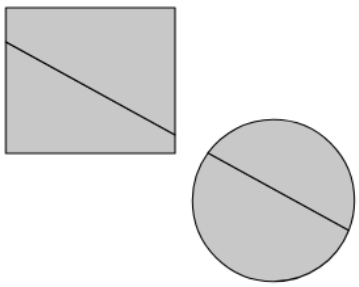


Figure 8.1 — The two parts produced by a straight cut, moved apart

8.2 Cutting with a curve

1. Select the shape.
2. Click Cut with a curve, or press Shift+K.
3. Click to place each node of the cutting curve, exactly as when drawing a curve.
4. Right-click or press Enter to perform the cut.

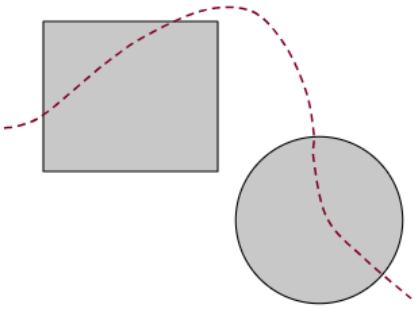


Figure 8.2.1 — The cutting curve drawn across the shape

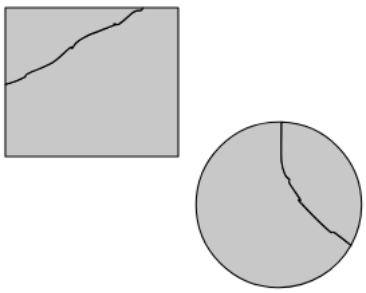


Figure 8.2.2 — The result, with the parts separated

8.3 Closing the cut edges

The Close the cut edges switch decides how the new edges are treated. With it off, the parts keep open contours and the cut edge has no outline. With it on, each part is closed along the cut, so the outline runs all the way round and a fill has a boundary to stop at.

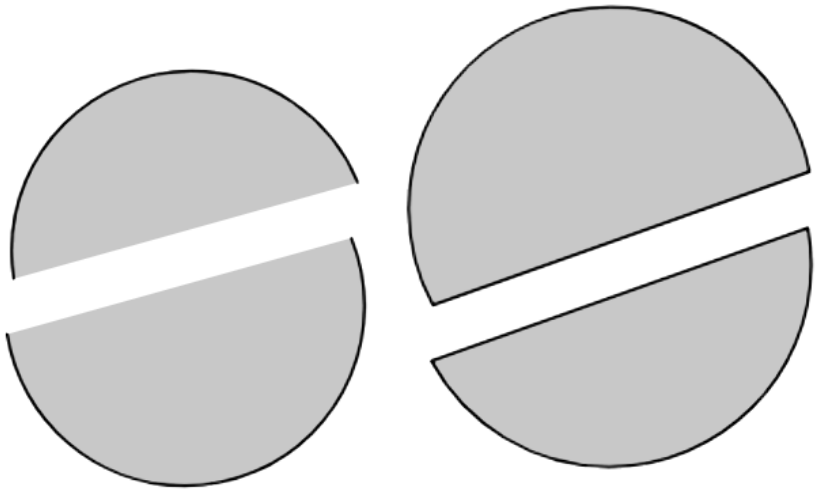


Figure 8.3 — The same cut with the closing switch off and on

9. Line anchors

An anchor ties the end of a line or curve to another shape. Once tied, the end follows that shape: move the shape and the line stretches after it, so a connector never comes adrift from what it points at.

9.1 Attaching an end

1. Draw a line or a curve so that one end lies on or near the shape it should follow.
2. Select the line.
3. Open the Link section of the properties panel.
4. Tick First node, or Last node, depending on which end you are tying.
5. Click the shape that the end should follow when the program asks you to pick a target.
6. Move the target shape — the end of the line follows it.

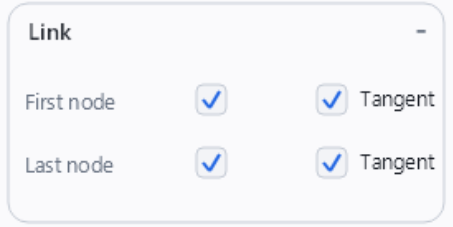


Figure 9.1 — The Link section

9.2 The Tangent switch

Each end has its own Tangent switch. With it off, the anchored end simply sits on the target. With it on, the line also leaves the target perpendicular to its edge, which is what makes a connector look deliberate rather than accidental.

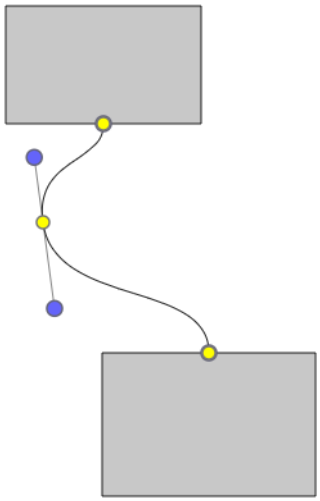


Figure 9.2 — A curve anchored to a rectangle, leaving the edge tangentially

9.3 Releasing an end

1. Select the line.
2. Open the Link section.
3. Untick First node or Last node. The end stays where it is but stops following the shape.

> *Deleting the target shape releases the anchor automatically; the line stays in place.*

10. Grouping

A group is a single object made of several shapes. Grouping is the usual way to keep an assembly together so it can be moved, scaled and copied as one.

10.1 Grouping

1. Select two or more shapes.
2. Press Ctrl+G, click Group on the object toolbar, or choose Group selected from the right-click menu.
3. The individual handles are replaced by one set around the whole group.

A group can contain other groups. Repeat the operation to build an assembly out of sub-assemblies.

10.2 Ungrouping

1. Select the group.
2. Press Ctrl+Shift+G, click Ungroup, or choose Ungroup selected from the right-click menu.
3. The members become independent shapes again and stay selected, so a second Ctrl+Shift+G breaks up any nested groups as well.

10.3 What a group does to properties

- Moving, rotating, mirroring and scaling apply to every member at once.
- The properties panel shows the properties of the group; changing the outline color with a group selected applies it to the members that have an outline.
- The stacking order inside a group is preserved. Bring to front and Send to back move the whole group relative to other objects on the page.

11. Rulers, guides, grid and snapping

11.1 Rulers

Press Ctrl+R, or the Rulers button, to show or hide the rulers. They read in the units of the page and start from the page zero point, so the numbers on the ruler match the X and Y in the properties panel.

11.2 Creating a guide

1. Make sure the rulers are visible — Ctrl+R.
2. Press the mouse button on the horizontal ruler and drag downwards onto the sheet. A horizontal guide follows the pointer.
3. Release where the guide should sit.
4. Drag from the vertical ruler in the same way for a vertical guide.

Guides are not printed and not exported. They exist to line work up.

11.3 Moving and removing a guide

1. Point at the guide — the pointer changes over it.
2. Drag it to a new position; the ruler shows where it is.
3. To remove it, drag it back onto the ruler it came from, or select it and press Del.
4. Ctrl+; hides all guides without deleting them.

11.4 Rotating a guide

1. Point at the guide — the pointer changes over it.
2. Drag the arrowed arc adjusting the guide angle.

11.5 The grid

Ctrl+#, or the Grid button on the document toolbar, shows the grid. The grid is drawn in three weights — ordinary lines, every fifth line and every tenth line — so that squares can be counted by eye. The axes through the page zero are darker still. The three colors and the axis color are set in Settings, on the Grid page.

11.6 Snapping

Snapping makes a dragged point jump to something meaningful instead of landing at an arbitrary place. Three switches on the main toolbar control it.

Switch	What it snaps to
Snap to objects	Points and edges of other shapes: corners, centres, midpoints, ends.
Guides	The guidelines you dragged out of the rulers.
Dynamic guides	Alignment lines that appear while dragging, showing when the shape lines up with a neighbor.

Snapping also targets the page origin and the grid. The order of preference is: page origin, then points of shapes, then dynamic guides, then edges, then the grid — so a corner of another shape always wins over a grid line.

> *Hold Ctrl while dragging to suspend snapping for that movement. Hold Alt while moving a shape to move it freely with no snapping at all.*

11.6 Snapping by a chosen node

In shape mode, moving the pointer over a shape highlights the nearest node with a small circle. If you start the drag on that highlighted node, the whole shape moves but only that node is tested against the snapping targets. This is how a specific corner is placed on a specific point. If the drag starts anywhere else, the node nearest to the pointer is used instead.

12. Custom shapes and .vdrs files

A custom shape is a drawing you saved once and can drop into any document afterwards — a valve, a title block, a company logo. Custom shapes live in the Objects menu, under Custom shapes.

12.1 Where custom shapes come from

The Custom shapes submenu gathers sets from three different places, and the difference matters.

Source	Where it is kept
This document	Sets created while this document was open. They travel inside the .vdr file, so a colleague who opens your drawing sees them too.
Library	Files in the Resources\Shapes folder beside the program. Available in every document, on this computer only. Sub-folders of Resources\Shapes appear as sub-menus.
Visio stencils	Loaded from a .vss or .vssx file for the current session only. They are not written to disk and not saved with the document.

12.2 Creating a custom shape

1. Draw the shapes you want to save, and select them all.
2. Open Objects → Custom shapes → Add selection to custom shape.
3. Type a name and confirm.
4. The set is written to Resources\Shapes as a .vdrs file and a copy is stored inside the current document.

The shapes are saved centred on their own middle, so that inserting the set later does not carry the position it happened to have when you saved it.

12.3 Inserting a custom shape

1. Open Objects → Custom shapes.
2. Find the set — under This document, under its library sub-menu, or under the name of a loaded stencil. Each entry shows a small picture of the shape.
3. Click the entry.
4. The shapes appear in the middle of the visible area, already selected, so the next drag moves them where you want.

The set is scaled to the units of the page it lands on. A set saved on a page in millimetres keeps its physical size when inserted into a page in inches.

12.4 Importing .vdrs files

A .vdrs file is one saved set. Import brings other people's sets into your library.

1. Open Objects → Custom shapes → Import custom shapes (*.vdrs).
2. Select one or several .vdrs files — the dialog allows a multiple selection.
3. Confirm.
4. The files are copied into Resources\Shapes, so the import survives the original files being moved or deleted.
5. The new sets appear in the Custom shapes menu straight away, at the level of the sub-folder they were copied into.

> *Nothing is reported when the import succeeds — the new entries in the menu are the confirmation. A message appears only if a file could not be read, and it names the file.*

Removing a set

1. Open Objects → Custom shapes and find the entry.
2. Right-click it.
3. Confirm the deletion. A library set is deleted from disk; a set belonging to the document is removed from the document only.

12.5 The same import from the toolbar

The Import button on the main toolbar (Ctrl+I) accepts .vdrs files as well. It sorts what you selected by type: .vdr files are merged into the drawing, .pdf files become pages of shapes, .vdrs files go into custom shapes and Visio files are handled as described in the next chapter. Files of several kinds can be selected together.

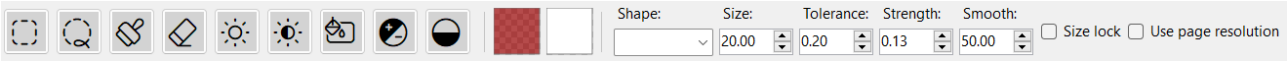
13. Image editing

An image placed on a page can be retouched inside VDraw. Switching to image mode replaces the object toolbar with brushes, selection tools and their settings.

13.1 Entering and leaving image mode

1. Select the image shape with the Select tool.
2. Press Ctrl+M, or click Image editing mode on the tool palette.
3. The image toolbar appears above the canvas. The window contents shift down to make room for it, and the pointer moves with them so that the shape under it does not jump.
4. Press Ctrl+M or Esc to leave image mode. The toolbar disappears and the shape behaves as an ordinary object again.

13.2 The image toolbar



Tool	What it does
Eraser	E — clears pixels to the background color.
Brush	B — paints with the foreground color. Soft edge, strength and smoothing all apply.
Rectangular selection	R — selects a rectangular area of pixels.
Lasso selection	L — selects a free-hand area. Shift adds to the selection, Alt subtracts.
Fill	G — fills an area of similar color. Tolerance decides how similar is similar enough.
Lighten	Brightens pixels under the brush.
Darken	Darkens pixels under the brush.
Contrast	Raises contrast under the brush.
Black and white	Removes color from pixels under the brush.
Foreground color	Drawing color for the brush and the fill.
Background color	What the eraser leaves behind.
Size	Tool size in pixels.
Tolerance	How far a pixel may differ from the one clicked and still be affected.
Strength	Strength of the effect per stroke.
Smooth	Edge softness of the stroke.
Shape	Brush tip shape — round or square.
Size lock	When enabled, the size of the tool does not depend on the view zoom.
Use page resolution	When enabled, the picture is rendered as per page resolution (dpi).

13.3 Retouching: a worked sequence

1. Select the image and press Ctrl+M.
2. Click the Brush tool.
3. Set Size to the width of stroke you need, and Smooth to about half of it for a soft edge.

4. Click the Foreground color button and pick the color.
5. Drag on the image to paint. Each stroke is one undo step.
6. Press Ctrl+Z to take back a stroke.

13.4 Working on part of the image

1. Click Rectangular selection, or Lasso selection.
2. Drag to enclose the area. A dashed outline marks it.
3. Hold Shift and drag again to add another area; hold Alt and drag to cut an area out of the selection.
4. Use any brush or effect — it now applies inside the selection only.
5. Press Ctrl+C to copy the selected pixels, or Del to clear them to the background color.
6. Click outside the selection to release it.

13.5 Filling an area

1. Click the Fill tool, or press G.
2. Set Tolerance: a low value fills only pixels almost identical to the one you click, a high value spreads across shades.
3. Click inside the area. Everything connected to the clicked pixel and close enough in color takes the foreground color.

14. The properties panel

The properties panel on the right shows everything about the current selection. It is divided into sections; a section can be collapsed by clicking its header, and the panel remembers which sections you left collapsed.

14.1 Several shapes at once

When more than one shape is selected the panel still shows every section. Two rules apply.

- A value that differs between the selected shapes is shown as an empty field. Typing into it gives every selected shape that value.
- A property applies only to the shapes that have it. Changing the radius with a circle and a text object selected changes the circle only; the text is left alone.

> *Layer is the exception: it is disabled when several shapes are selected, because the layer is exactly what puts one shape above another and they cannot share a value.*

14.2 General



Figure 15.1 — The General section

Property	Meaning
Name	The name of the shape, shown in lists and menus. Leaving it empty keeps the previous name.
X, Y	Position of the left and top edge of the shape, measured from the page zero.
Width, Height	Size of the shape. Each shape is resized about its own centre.
Rotation	Angle in degrees.
Layer	Stacking order. A higher number is drawn over a lower one.

Some shapes add their own fields under these. A rounded rectangle adds Round X and Round Y, a chamfered rectangle Chamfer X and Chamfer Y, a circle Radius, an ellipse Radius X and Radius Y, and a regular polygon Sides together with its two radii.

14.3 Line

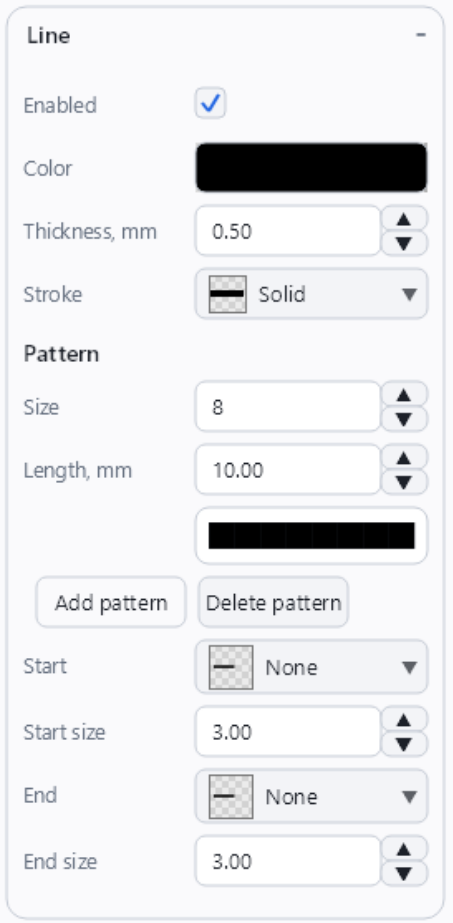


Figure 15.2 — The Line section

Property	Meaning
Enabled	Draws the outline or leaves it out.
Color	Outline color. Click the swatch to open the color dialog.
Thickness	Line width, in the units of the page.
Stroke	Solid, dashed, dotted and the other ready-made styles.
Pattern — Size	Width of the dash pattern grid, in cells.
Pattern — Length	Length of one full cycle of the pattern, in the units of the page.
Pattern grid	Draw your own dash pattern by clicking the cells.
Add pattern / Delete pattern	Saves the pattern you drew into the list of stroke styles, or removes it.
Start, End	Arrowheads and other glyphs at the two ends of the line.
Start size, End size	Size of those glyphs, as a multiple of the line width.

14.4 Fill

The Fill section starts with an Enabled switch and a row of buttons choosing the type of fill. The rest of the section changes with the type.

None



Figure 15.3 — No fill

The shape has an outline only. Clicks pass through the middle of it.

Solid

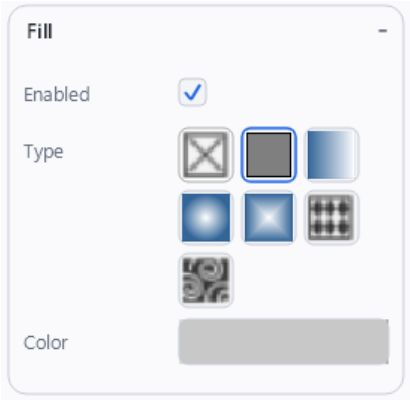


Figure 15.4 — Solid fill

One color, chosen with the swatch.

Linear gradient

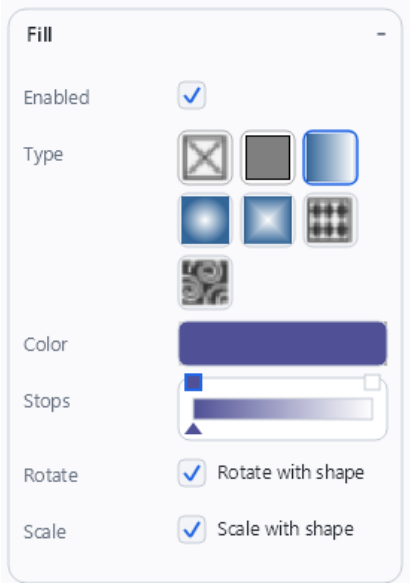


Figure 15.5 — Linear gradient

Color changes along a straight line. The stops below the preview set the colors; drag the handles on the shape itself to set the direction and length of the gradient. Rotate with shape and Scale with shape decide whether the gradient follows the shape when it is turned or resized.

Radial and rectangular gradients

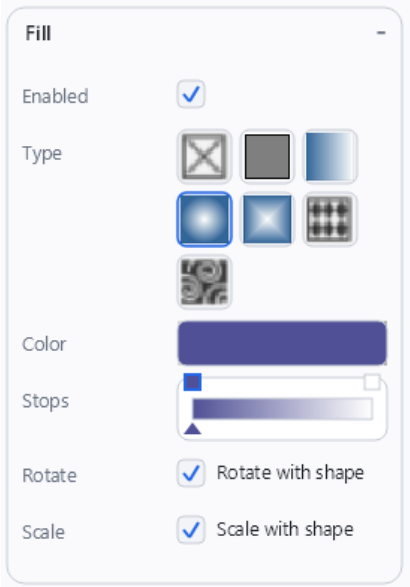


Figure 15.6 — Radial gradient

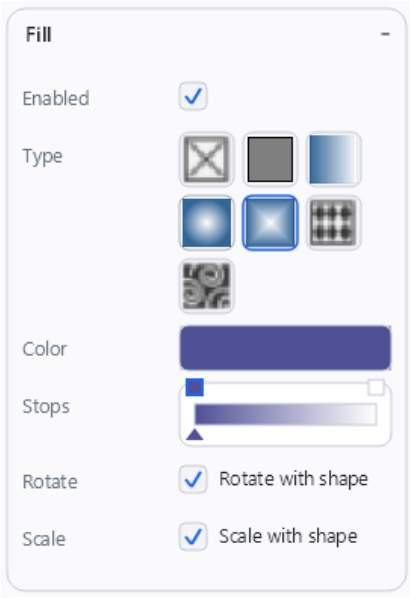


Figure 15.7 — Rectangular gradient

The same stops, spread from a centre outwards in a circle or in a rectangle.

Pattern

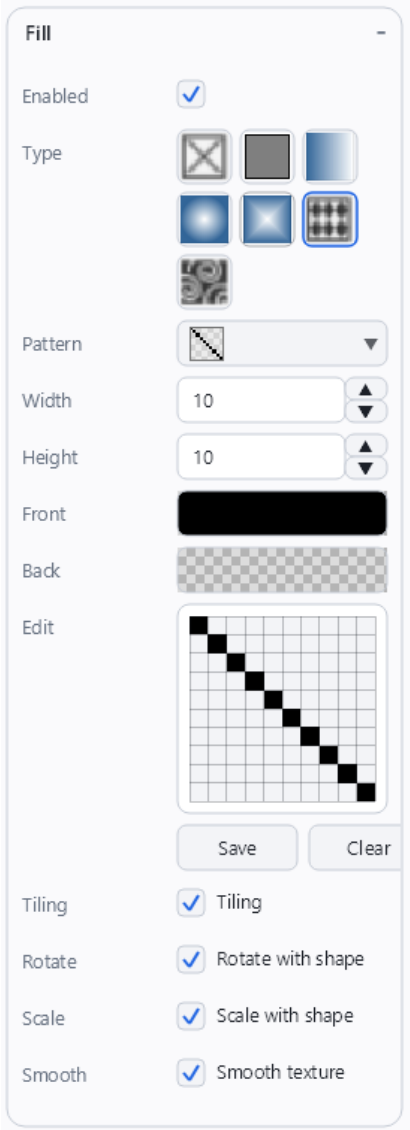


Figure 15.8 — Pattern fill

A small drawing repeated over the shape. Choose a ready pattern from the list, or draw your own in the grid: click a cell to set it in the foreground color, click again to clear it. Width and Height set the size of the tile in cells, and the two color buttons set the foreground and the background. Tiling repeats the tile; with it off the tile is drawn once. Save pattern adds your drawing to the list for later use.

Texture

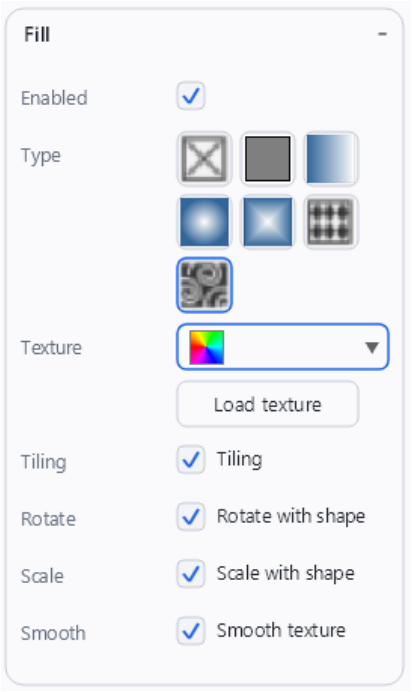


Figure 15.9 — Texture fill

A picture used as a fill. Load texture reads an image file. Smooth decides whether the picture is interpolated when it is stretched; with it off the pixels stay crisp, which is what you want for a small pattern and not what you want for a photograph.

14.5 Shadow

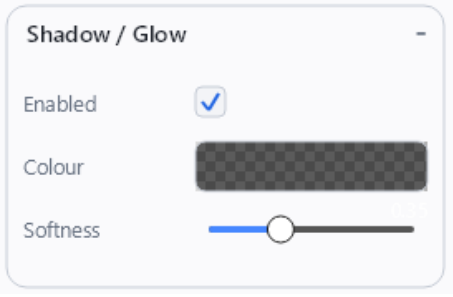


Figure 15.10 — The Shadow section

Enabled turns the shadow on. Color sets it, and Softness blurs the edge — nought is a hard shadow, one is a wide diffuse glow. The offset is set by dragging the shadow handles on the shape itself.

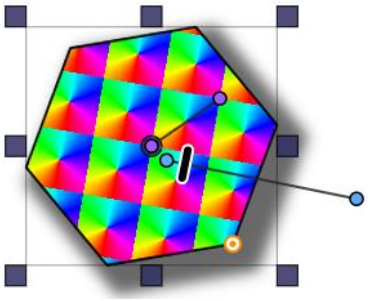


Figure 15.11 — A shape with a shadow

14.6 Transparency

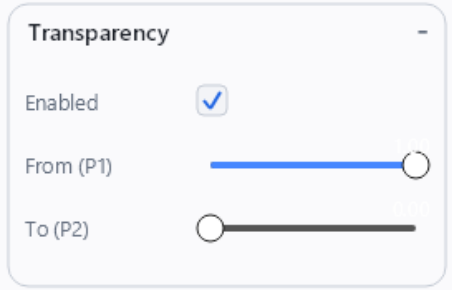


Figure 15.12 — The Transparency section

Transparency fades the shape out along a line. Start and End set how opaque the shape is at the two ends of that line, and the line itself is dragged on the shape with the two handles.

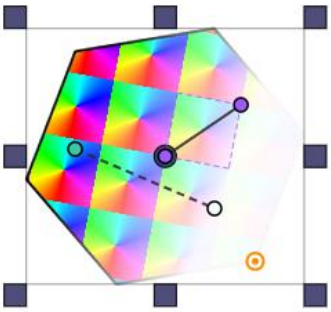


Figure 15.13 — A shape fading out

14.7 Text

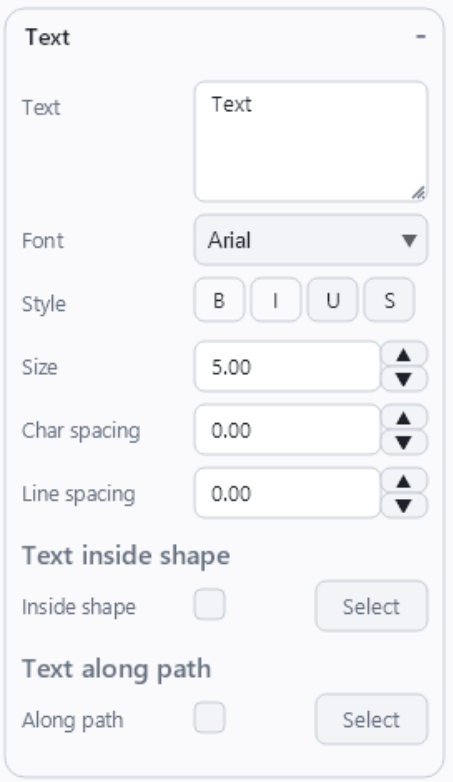


Figure 15.14 — The Text section

Property	Meaning
Text	The text itself.
Font	Typeface.
Size	Type size, in the units of the page.
Bold, Italic, Underline, Strikethrough	Type style.
Character spacing	Extra space between letters.
Line spacing	Extra space between lines.
Factor	For dimensions: multiplies the measured value before it is written into the label. Use it when the drawing is at a scale.
Value	For dimensions: the number written in the label. Type your own to label a drawing that is not to scale; the field returns to the measured value as soon as the dimension is stretched or the shapes it is attached to move.

14.8 Link

Described in chapter 9.

15. Export and printing

15.1 Exporting a raster image

1. Select the shapes to export. Nothing selected means nothing to export.
2. Press Ctrl+E, or click Export selection on the main toolbar.
3. Choose the file name and the format — PNG, JPEG, BMP or SVG.
4. For a raster format the settings dialog opens.

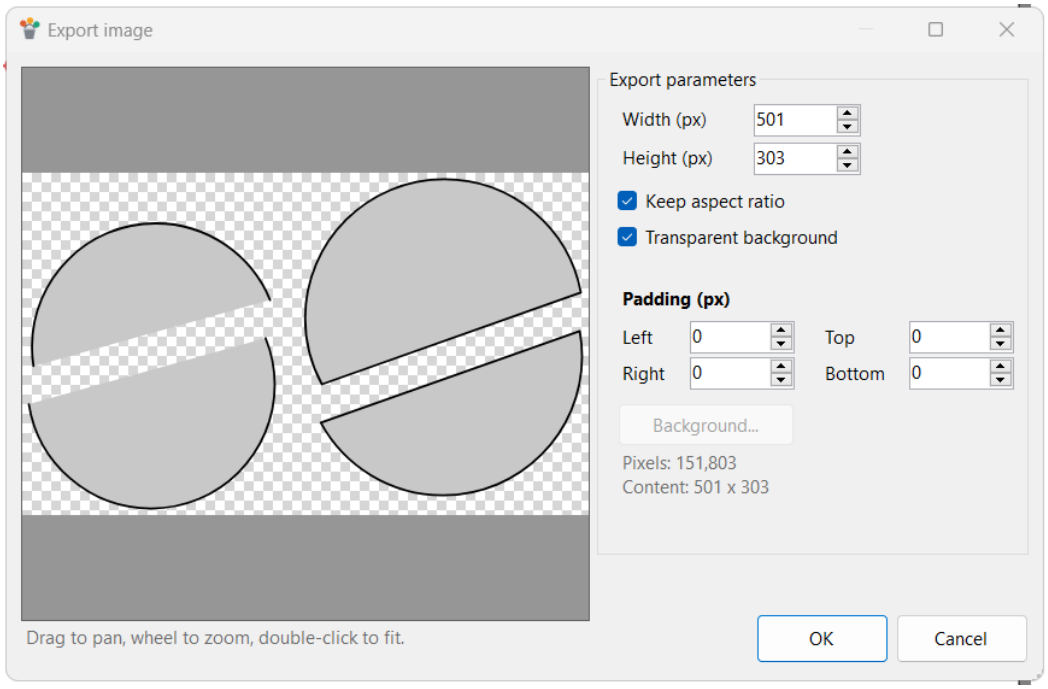


Figure 16.1 — The raster export dialog

Control	Meaning
Width, Height	Size of the file in pixels.
Keep aspect ratio	Changing one side changes the other proportionally.
Transparent background	Available for PNG only, because JPEG and BMP cannot store transparency.
Padding	Empty margins inside the image, in pixels. The file keeps the size you asked for; the drawing is squeezed to make room.
Background	The color behind the drawing when transparency is off.
Preview	Shows exactly what will be written. Drag to pan, wheel to zoom, double-click to fit.

> *SVG is a vector format and has no such dialog — the shapes are written as shapes.*

15.2 Exporting an animation

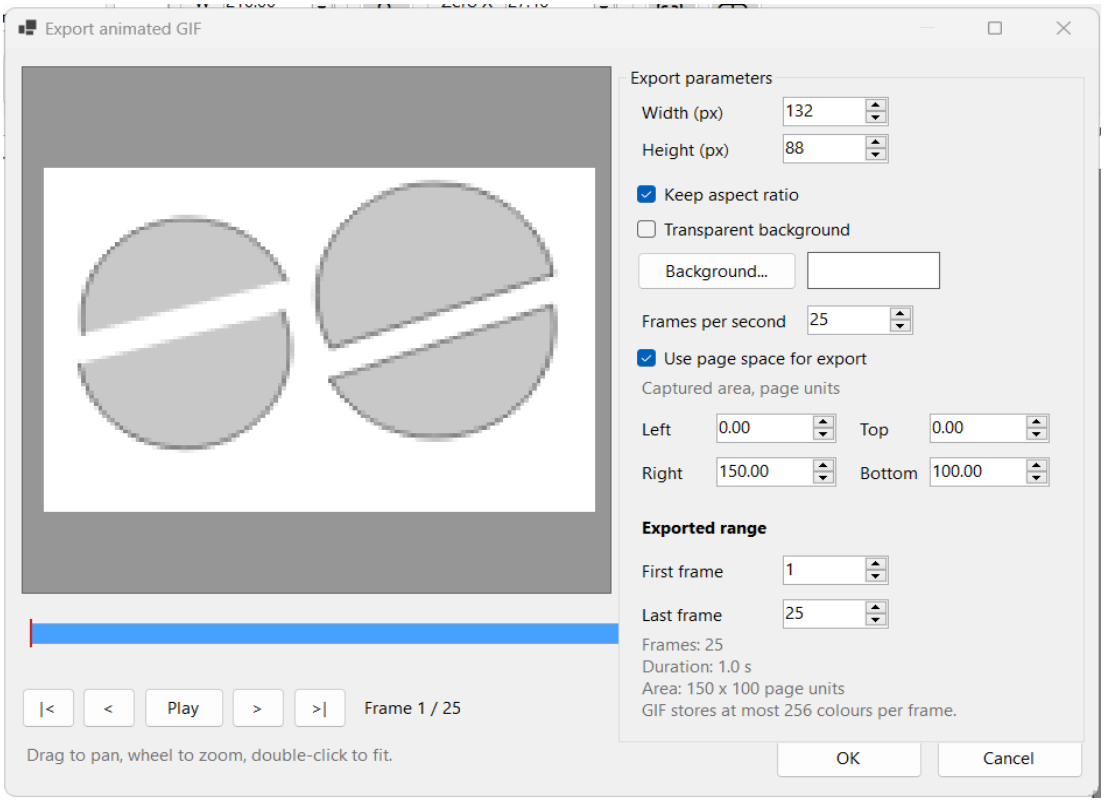


Figure 16.2 — The GIF export dialog

The animation bar at the bottom of the window sets keys on frames; GIF export writes the result as an animated file. Frames, frames per second and the area to capture are set in the dialog. GIF stores at most 256 colors per frame, so gradients may band.

15.3 Printing

1. Press Ctrl+P, or click Print.
2. Choose the printer and its options.
3. Use Printable area borders on the document toolbar beforehand to see the margins your printer cannot reach — anything outside them will be clipped.

16. Animation

VDraw animates a drawing by recording the state of shapes on numbered frames. You put a shape where it should be at frame 1 and record a key, move to frame 25, move the shape and record another key, and the program fills in everything between the two. Nothing else has to be prepared: any shape already on the page can be animated, and any property the properties panel can change can be animated too.

Everything to do with animation lives in the panel along the bottom of the window.

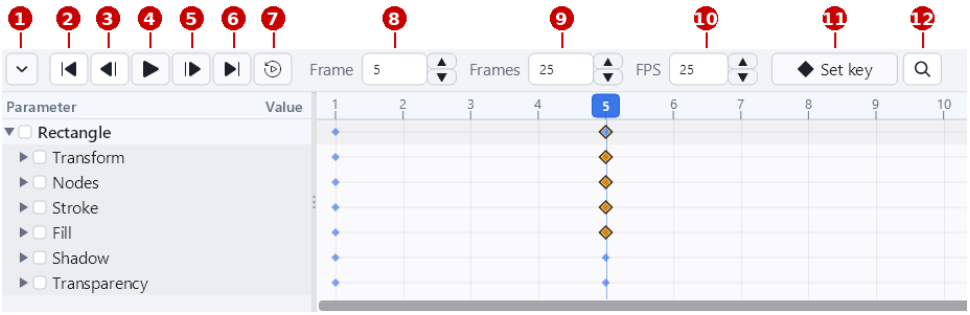


Figure 16.1 — The animation panel: toolbar (1–12), parameter tree on the left, timeline on the right

16.1 Making your first animation

This sequence produces a rectangle that slides across the page in one second. Read it once through; the rest of the chapter explains each piece in detail.

1. Draw a rectangle on the page and leave it selected. The parameter tree at the bottom-left immediately shows its name with the groups Transform, Nodes, Stroke, Fill, Shadow and Transparency underneath.
2. Type 25 into the Frames box (9) — the animation will be 25 frames long.
3. Leave FPS (10) at 25, so those 25 frames play in one second.
4. Type 1 into the Frame box (8), or press Home. The playhead moves to the first frame.
5. Click Set key (11). Diamonds appear at frame 1 on every row of the tree: the state of the rectangle is now recorded there.
6. Type 25 into the Frame box, or press End. The playhead moves to the last frame.
7. Drag the rectangle to where it should finish.
8. Click Set key again. A second column of diamonds appears at frame 25.
9. Click Play (4). The rectangle slides from the first position to the second and stops.
10. Click Play again to stop playback.

> *Recording a key does not lock the shape. You can keep drawing, moving and restyling at any frame; what is recorded is only what you asked to record, when you asked.*

16.2 The toolbar control by control

The numbers refer to Figure 16.1.

No. / control	What it does
1 — Collapse	Folds the panel down to its toolbar and unfolds it again. Use it to give the canvas the whole window; the animation itself is not affected.
2 — To start	Jumps the playhead to frame 1. Same as pressing Home.
3 — Step back	Moves the playhead one frame earlier. Same as the Left arrow key.

No. / control	What it does
4 — Play / Pause	Starts playback from the current frame and turns into a pause button. Same as the Space bar. Playback loops back to the beginning when it reaches the end.
5 — Step forward	Moves the playhead one frame later. Same as the Right arrow key.
6 — To end	Jumps to the last frame. Same as pressing End.
7 — Reverse	A switch, not a command. While it is pressed in, playback runs from the current frame backwards. Nothing happens until you press Play.
8 — Frame	The frame the playhead is on, and therefore the state of the drawing you are looking at. Type a number, use the arrows, or click on the ruler in the timeline. Whatever you draw or change is applied at this frame.
9 — Frames	The total length of the animation, from 1 to 10 000 frames. Shortening it does not delete keys that fall beyond the new end; they simply stop being reached until you lengthen the animation again.
10 — FPS	Frames per second, from 1 to 120, twenty-five by default. It decides how fast the animation plays and how long the exported file lasts: 25 frames at 25 FPS is one second, the same 25 frames at 10 FPS is two and a half.
11 — Set key	Records the current state of the selected shapes at the current frame. Which properties are recorded depends on the checkboxes in the tree — see 16.4.
12 — Zoom	A switch. While it is pressed in, dragging across the ruler or the timeline selects a range of frames and the view zooms to it. Press it again, or finish the drag, to return to normal editing.

16.3 The parameter tree

The left half of the panel lists what can be animated. The top row is the shape itself, named as in the General section of the properties panel; under it are the groups of its properties. Selecting a different shape on the page changes the tree; selecting several shows all of them, one root row each.

Group	What it holds
Transform	Position and shape of the object as a whole: X, Y, Scale X, Scale Y, Skew X, Skew Y and Visible.
Nodes	The geometry itself. For a rectangle this is its rectangle and, where applicable, the corner radius; for a circle, the centre and radius; for a curve, every node and its control points.
Stroke	Outline colour, width, dash length and the rest of the Line section.
Fill	Fill colour, gradient stops, pattern size and so on.
Shadow	Colour, offset and softness.
Transparency	Start and end opacity and the direction of the fade.

- The triangle at the left of a row opens and closes the group.
- The checkbox beside it selects the row for recording — see the next section.
- The Value column shows what the parameter is at the current frame, so you can read off an interpolated value without leaving the panel.

> *Position is animated as X and Y of the shape's transform, and size as Scale X and Scale Y — the same six numbers a transform matrix is made of. This is why an animated shape scaled to zero disappears and reappears cleanly, while a shape animated by its Width would not.*

16.4 Choosing what Set key records

The checkboxes decide the scope of the Set key button, and the rule has two halves.

- With nothing checked anywhere in the tree, Set key records every parameter of every selected shape. This is the usual way to work: it guarantees that the frame is captured completely.
- With at least one checkbox ticked, Set key records only the ticked parameters. Everything else is left as it is.

Tick a group and every parameter inside it is ticked with it. This is how you animate one aspect of a shape in isolation — for example, tick Fill only, and the colour changes over time while the position, recorded elsewhere, is left alone.

1. Open the group you want by clicking its triangle.
2. Tick the checkbox of the group, or of the individual parameters inside it.
3. Move to the frame you want.
4. Change the shape.
5. Click Set key — only the ticked parameters get a key.

16.5 Keys

A key is one recorded value of one parameter at one frame. On the timeline a key is drawn as a diamond on the row of its parameter.

What the diamonds mean

- An orange diamond on a parameter row is a real key: that parameter has a recorded value at that frame.
- A diamond on a group row, or on the shape row at the top, is a summary. It says that something inside that group has a key on that frame. It is not itself a key, and it never creates one.
- A small blue dot on a row marks a frame the parameter passes through with no key of its own — the value there is calculated, not recorded.

Recording a key

1. Set the current frame.
2. Arrange the shape as it should look on that frame.
3. Click Set key, which records the whole selection, or double-click on an empty spot in one parameter's row, which records only that parameter at the frame you double-clicked.

Recording a key where one already exists overwrites it. This is how a key is corrected: go to the frame, fix the shape, press Set key again.

What happens between keys

Between two keys the value is interpolated, and the kind of value decides how.

Kind of parameter	Behaviour between keys
Numbers — position, size, width, opacity	Interpolated linearly and evenly.
Colours	Interpolated through red, green, blue and alpha separately, so a colour changes smoothly from one to the other.
Points — node positions, gradient handles	Both coordinates interpolated, so the point travels in a straight line.
Switches — Visible, Enabled	Held. The value of the earlier key stays in force until the next key, then changes at once. A switch is never half on.

Before the first key and after the last one the value simply stays at that key. A parameter with a single key therefore holds that value for the whole animation, and a parameter with no keys at all is not animated: it keeps whatever the drawing has.

16.6 Selecting keys

Action	Result
Click a diamond	Selects it and deselects everything else.
Click a diamond on a group row	Selects every key of that group standing on that frame — the whole column at once.
Shift + click a diamond	Adds it to the selection. Shift-clicking a diamond that is already fully selected removes it again.
Drag on empty timeline	Draws a frame around the keys and selects those inside it.
Shift + drag	Adds the framed keys to the existing selection.
Click empty timeline without dragging	Clears the selection and moves the playhead to that frame.

16.7 Moving and deleting keys

Moving

1. Select the keys you want to move.
2. Press the mouse button on any of the selected diamonds.
3. Drag left or right. All selected keys move together, keeping the spacing between them.
4. Release. The whole move is one undo step.

Moving keys is how the timing of an animation is adjusted without touching the drawing. Dragging the last column of keys further to the right makes the motion slower; dragging it left makes it quicker.

Deleting

1. Select the keys.
2. Press Del.

Deleting the keys of a parameter down to one leaves that parameter holding a constant value; deleting the last one stops it being animated at all.

> *Every operation on keys — recording, overwriting, moving, deleting — is a normal undo step. Ctrl+Z takes back the whole operation, including a move of fifty keys at once.*

16.8 Moving around the timeline

Action	Result
Click on the ruler	Moves the playhead to that frame.
Drag the playhead handle	Scrubs: the drawing updates continuously as you drag, so motion can be checked by hand.
Drag the ruler away from the handle	Pans the visible range of frames.
Mouse wheel over the timeline	Zooms the frame scale in and out, centred on the pointer.
Ctrl + wheel over the timeline	Scrolls the rows instead of zooming.
Wheel over the parameter tree	Scrolls the rows.

Action	Result
Zoom switch (12) + drag	Zooms to exactly the range of frames you drag over.
Drag the divider between the tree and the timeline	Changes the width of the tree.
Drag the top edge of the panel	Changes the height of the panel.

16.9 Keyboard shortcuts in the panel

These work when the animation panel has the focus — click anywhere in it first.

Key	Command
Space	Play / pause.
Left / Right	One frame back / forward.
Home / End	First frame / last frame.
Del	Delete the selected keys.
Shift + click, Shift + drag	Add to the key selection.
Ctrl + wheel	Scroll the rows instead of zooming.

16.10 A worked example: a bouncing ball

A short exercise that uses keys, moving them, and two different groups of parameters.

1. Draw a circle near the top of the page and leave it selected.
2. Set Frames to 20 and FPS to 25.
3. Go to frame 1 and click Set key.
4. Go to frame 10, drag the circle down to the floor, and click Set key. Play: the ball falls.
5. Go to frame 20, drag the circle back up to the top, and click Set key. Play: the ball falls and rises.
6. Now flatten it on impact. Open Transform in the tree and tick Scale X and Scale Y.
7. Go to frame 10 and squash the circle a little with its handles — wider and lower.
8. Click Set key. Because two parameters are ticked, only the scale is recorded; the positions you set earlier are untouched.
9. Play. The ball squashes as it lands and returns to its shape as it rises.
10. If the fall looks too slow, select the column of diamonds at frame 10 by clicking the diamond on the shape row, and drag it left to frame 6. The fall becomes quicker and the rise slower.

16.11 Saving and exporting

- Keys are part of the document. Saving the .vdr file saves the animation with it; no separate step is needed.
- Each page carries its own animation, with its own length and frame rate.
- To produce a file others can watch, use the GIF export. It writes the frames as an animated image, using the Frames and FPS values set here.
- Ordinary raster export writes the drawing as it stands at the current frame, which is a convenient way to pull a still out of an animation: move the playhead to the frame you want and export the selection.

17. Settings

The Settings dialog holds the preferences of the program itself — everything about how the editor looks and behaves, as opposed to what is inside your drawing. The settings are saved when you press OK or Apply and are restored the next time the program starts.

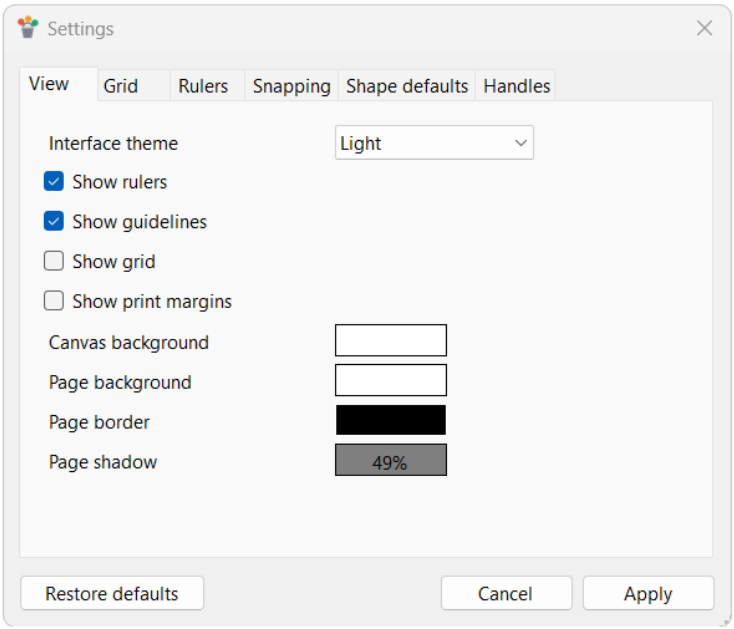


Figure 17.1 — The Settings dialog

17.1 The pages

Page	Contents
View	Interface theme, the switches for rulers, guides, grid and print margins, and the colors of the canvas, the page, its border and its shadow.
Grid	Colors of the ordinary, fifth and tenth grid lines, of the axes through the page zero, and of the printable area border.
Rulers	Thickness of the rulers and the length of the origin, major and minor marks.
Snapping	The three snapping switches, the snap distance in pixels, and the size and colors of the snap markers.
Shape defaults	Outline color and width, dash cycle length and fill color given to newly created shapes.
Handles	Size and colors of the selection handles, the grab radius, and the size of curve nodes and control points.

17.2 The buttons

- Apply — writes the settings and leaves the dialog open, so the effect can be judged.
- OK — writes them and closes.
- Cancel — closes without writing. Anything already applied stays applied.
- Restore defaults — returns every setting to the value the program shipped with, not to the value it had when the dialog was opened. Nothing is written until Apply or OK.

18. Keyboard shortcuts

Shortcut	Command
Ctrl+N / Ctrl+O	New document / Open document
Ctrl+S / Ctrl+Shift+S	Save / Save as
Ctrl+I / Ctrl+E / Ctrl+P	Import / Export selection / Print
Ctrl+Z / Ctrl+Y	Undo / Redo
Ctrl+C / Ctrl+V / Del	Copy / Paste / Delete selected
Esc	Select tool
L / B / R / C / E	Lines / Curve / Rectangle / Circle / Ellipse
P / Shift+P / T	Regular polygon / Free polygon / Text
Ctrl+Shift+I / M	Image / Measure distance
Ctrl+G / Ctrl+Shift+G	Group / Ungroup
Ctrl+Shift+Up / Ctrl+Shift+Down	Bring to front / Send to back
Ctrl+Left / Ctrl+Right	Rotate 90° counter-clockwise / clockwise
Shift+H / Shift+V	Flip horizontally / vertically
Ctrl+Alt+U / S / I / X / C	Union / Subtract / Intersect / Exclude / Combine
Ctrl+Q / N	Convert to curve / Edit nodes
1 / 2 / 3	Straight / Smooth / Cusp node, while editing nodes
Ctrl+J / Ctrl+Shift+J	Merge nodes / Split node
K / Shift+K	Cut with a straight line / with a curve
Ctrl+R / Ctrl+; / Ctrl+#	Rulers / Guides / Grid
Ctrl+Shift+N	Snap to objects
Ctrl+M	Image editing mode
Ctrl+0 / Ctrl+1 / Ctrl+Shift+0	Zoom to all / Fit page / Zoom to selection
Ctrl + wheel	Zoom at the pointer
Ctrl (while dragging)	Suspend snapping
Alt (while moving)	Move with no snapping
F1	Help

19. If something goes wrong

VDraw keeps a log of errors. If the program behaves strangely, the log is the first place to look and the thing to send when reporting a problem.

- The file is Documents\VDraw\Logs\vdraw-errors.log.
- Every entry carries the date and time, the error itself, and the state of the program at that moment: version, system, memory, the number of pages and shapes, the active mode.
- The name of your document is never written to the log.
- When an error interrupts what you were doing, a message appears once per session for that particular error, with the path to the log.

19.1 Common situations

Symptom	What to check
An imported Visio drawing has flat curves	Splines are approximated by their end points. Nothing is wrong with the file.
Imported shapes are white	Their color comes from a Visio theme that is not part of the file. See 13.4.
A dimension shows the wrong number	Check Factor in the Text section, and whether Value has been overridden.
A shape cannot be node-edited	Primitives have no nodes until Convert to curve (Ctrl+Q) is applied.
Dragging ignores snapping	Check the three snapping switches, and whether Ctrl or Alt is being held.
Settings are back to defaults	The settings file could not be read; the reason is in the error log.